Name : Kanishka sharma , , Roll No : 26 , Computer Vision , Labsheet - 4

```
import cv2 import
numpy as np
import matplotlib.pyplot as plt

# Import libraries
print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

```
# Load PNG image
image_path = "ex4.png"

img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

if img is None:      print(" Image not
found") else:     print(" Image loaded
successfully")

    plt.figure(figsize=(6, 6))
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")     plt.show()
```

```
 Image loaded successfully
```

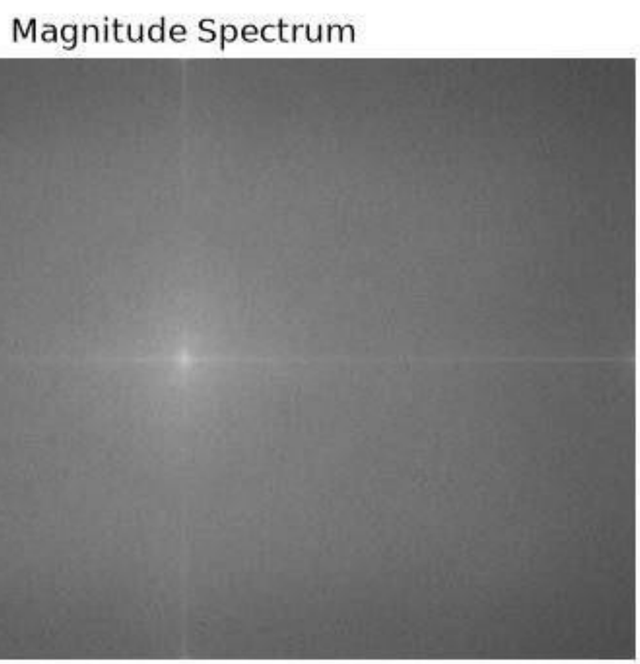
```python
# Calculate DFT
dft = cv2.dft(np.float32(img), flags=cv2.DFT_COMPLEX_OUTPUT)
print("DFT calculated successfully.")
```

```
DFT calculated successfully.
```

```python
# Shift zero frequency to center
dft_shift = np.fft.fftshift(dft)
print("Frequency shifted to center.")
```

```
Frequency shifted to center.
```

```python
# Calculate magnitude spectrum
magnitude = cv2.magnitude(
dft_shift[:, :, 0],
dft_shift[:, :, 1]
)

magnitude_spectrum = 20 * np.log(magnitude + 1)

plt.figure(figsize=(6, 6))
plt.imshow(magnitude_spectrum, cmap="gray")
plt.title("Magnitude Spectrum")
plt.axis("off") plt.show()
```



```python
# Compare image and spectrum
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(magnitude_spectrum, cmap="gray")
plt.title("Frequency Spectrum")
plt.axis("off") plt.show()
```

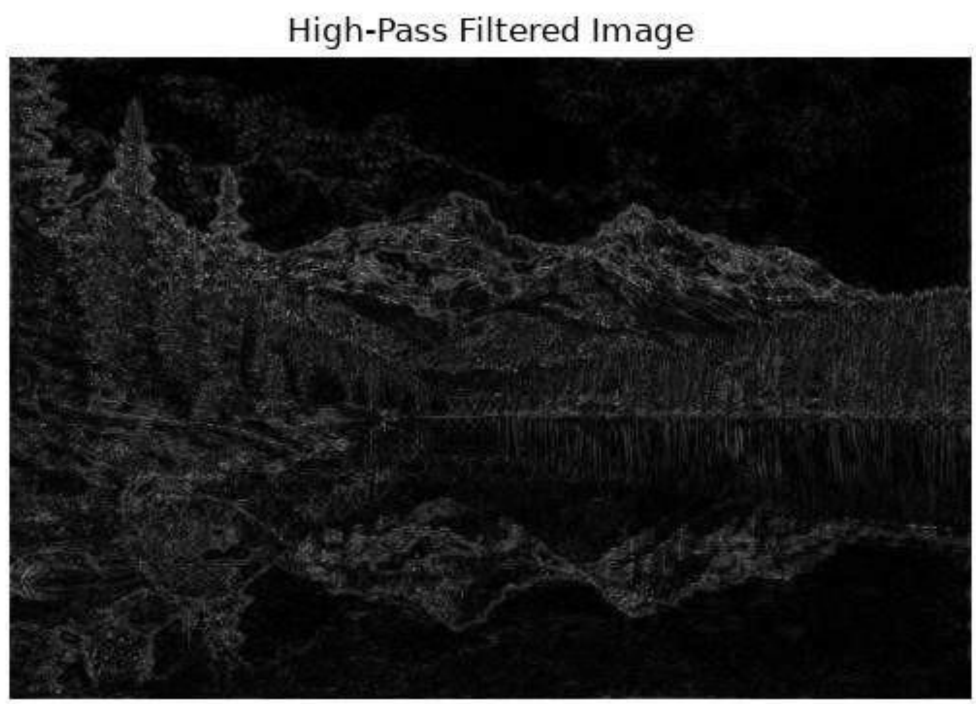

```python
# Create Low-Pass Filter
rows, cols = img.shape
crow, ccol = rows // 2, cols // 2

radius = 50

mask_low = np.zeros((rows, cols, 2), np.uint8)

cv2.circle(
mask_low,
    (ccol, crow),
radius,     (1,
1),
    -1
)

print("Low-Pass Filter created.")
Low-Pass Filter created.
# Apply Low-Pass Filter
filtered_low = dft_shift * mask_low
```

```
print("Low frequencies retained.")
Low frequencies retained.
# Shift back
f_ishift_low = np.fft.ifftshift(filtered_low)

# Apply inverse DFT
img_low = cv2.idft(f_ishift_low)

img_low = cv2.magnitude(
img_low[:, :, 0],
img_low[:, :, 1]
)

# Normalize image
img_low = cv2.normalize(
    img_low,
    None,
    0,
255,
    cv2.NORM_MINMAX
).astype(np.uint8)

plt.imshow(img_low, cmap="gray")
plt.title("Low-Pass Filtered Image")
plt.axis("off") plt.show()
```

Low-Pass Filtered Image

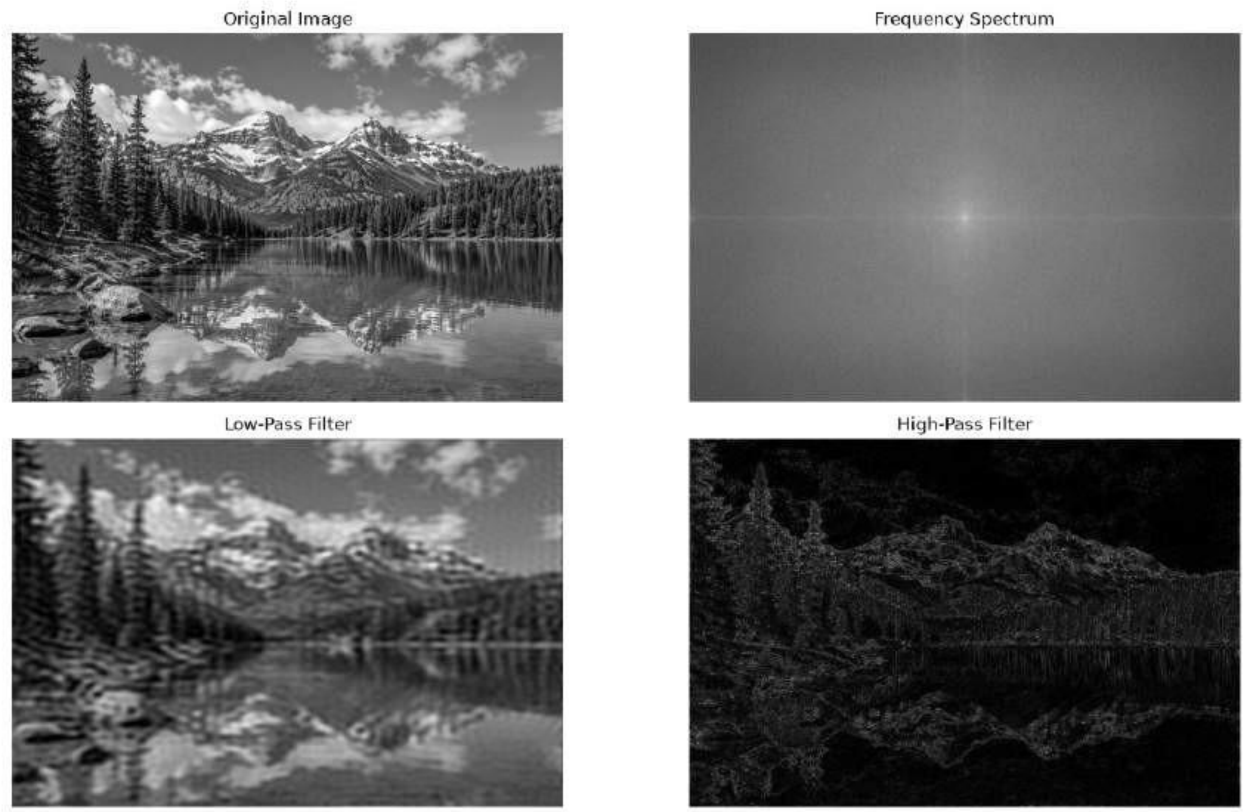
```
# Create High-Pass Filter
mask_high = np.ones((rows, cols, 2), np.uint8)

cv2.circle(
mask_high,
    (ccol, crow),
radius,     (0,
0),
    -1
)

print("High-Pass Filter created.")
High-Pass Filter created.
# Apply High-Pass Filter
filtered_high = dft_shift * mask_high
print("High frequencies retained.")
High frequencies retained.
# Shift back
f_ishift_high = np.fft.ifftshift(filtered_high)

# Apply inverse DFT
img_high = cv2.idft(f_ishift_high)
img_high = cv2.magnitude(
```

```
    img_high[:, :, 0],
img_high[:, :, 1]
)

# Normalize image
img_high = cv2.normalize(
    img_high,
    None,
    0,
255,
    cv2.NORM_MINMAX
).astype(np.uint8)

plt.imshow(img_high, cmap="gray")
plt.title("High-Pass Filtered Image")
plt.axis("off") plt.show()
```




```
# Compare all results
plt.figure(figsize=(14, 8))

plt.subplot(2, 2, 1)
plt.imshow(img, cmap="gray")
plt.title("Original Image")
plt.axis("off") plt.subplot(2, 2,
2) plt.imshow(magnitude_spectrum,
cmap="gray") plt.title("Frequency
Spectrum") plt.axis("off")
```

```
plt.subplot(2, 2, 3)
plt.imshow(img_low, cmap="gray")
plt.title("Low-Pass Filter")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(img_high, cmap="gray")
plt.title("High-Pass Filter")
plt.axis("off")

plt.tight_layout() plt.show()
```




```
# Save filtered images
cv2.imwrite("low_pass.jpg", img_low)
cv2.imwrite("high_pass.jpg", img_high)
print("Output images saved successfully.")
Output images saved successfully.
```

```python
# Display observations
print("Observations:")
print("1. Low-Pass Filter makes the image smoother.") print("2.
High-Pass Filter enhances edges and fine details.") print("3.
Frequency spectrum shows image frequency components.") print("4.
Fourier Transform is useful for image enhancement and
restoration.")
```

```
Observations:

1. Low-Pass Filter makes the image smoother.

2. High-Pass Filter enhances edges and fine details.

3. Frequency spectrum shows image frequency components.

4. Fourier Transform is useful for image enhancement and restoration.
```